

Huang Lab Coding Assessment

Arielle S Joasil

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1 Research Objective

Research Question: Can the imaging phenotype identify Homologous Recombination Deficiency (HRD)?

Biological Rationale: HR-deficient breast cancers exhibit distinct clinical and pathological characteristics, including aggressive angiogenesis, high proliferation rates, and necrosis—phenotypes that plausibly imprint on the enhancement kinetics and morphology observed in DCE-MRI. Using the Duke breast-MRI cohort, this study evaluates whether current imaging phenotypes carry detectable signals for HRD-enriched subtypes, utilizing Triple-Negative Breast Cancer (TNBC) as a biological proxy. This work aims to identify the specific radiomic features driving this classification and assess their interpretability within the context of tumor biology. This question was selected to leverage my specific expertise in constructing robust machine learning pipelines, bridging the gap between high-dimensional clinical data and predictive modeling.

2 Methodological Framework

2.1 Technical Environment and AI-Assisted Workflow

Computational Workflow: Due to temporary restrictions in GCP Service Account authentication during the active assessment window, the workflow was pivoted from a local CLI agent to a direct LLM-assisted implementation (Gemini Pro 3.1). This transition allowed for uninterrupted boilerplate generation while maintaining the structural integrity of the pipeline. Detailed implementation logs and milestone tracking are documented in `Implementation_Plan.md` and `Implementation_Plan_Milestones.md`.

To maximize productivity and ensure technical rigor within the assessment window, this project utilized an LLM-based agent (Gemini) as a pair-programming and documentation collaborator. Specifically, the agent was employed for:

- **Boilerplate Generation:** Automating the syntax for data processing pipelines and visualization routines.

- **Drafting and Refinement:** Iterative drafting of technical documentation and report structuring.

All architectural decisions, scientific interpretations, statistical validation, and biological conclusions were independently evaluated and verified by the author to ensure research integrity.

2.2 Data Acquisition and Preprocessing

Data were aggregated from the Duke Breast Cancer archive. The dataset contains 4 molecular subtypes, which were binarized to create a target variable (`HRD_Proxy`). Subtype 3 (TNBC) was encoded as the positive class (1), while all other subtypes were encoded as the negative class (0). To ensure data quality, entries containing missing values (NaNs) were excluded.

2.3 Exploratory Data Analysis

Radiomic features often exhibit significant collinearity, which can artificially inflate feature importance rankings. To mitigate this, a correlation threshold of $|\rho| \geq 0.8$ was applied, resulting in the removal of 382 redundant features and the retention of 147 informative features. The resulting correlation matrix (Fig. 1) confirms reduced redundancy in the feature space. Initial unsupervised clustering demonstrated that the dataset is highly imbalanced, with the positive target class (`HRD_Proxy` = 1) comprising only 18.35% of the cohort.

2.4 Modeling Strategy

A 10-fold, stratified cross-validation framework (90% train, 10% test) was implemented to maintain class distribution across splits and ensure model generalizability. Feature selection was driven by Gini importance calculated via a Random Forest Classifier.

Model robustness was validated through three approaches: (1) Feature space visualization using 3D Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP). (2) Visual validation of subgroups via clustered heatmaps. (3) Quantitative assessment of classification performance using Sensitivity, Specificity, Matthew’s Correlation Coefficient (MCC), and Accuracy. Given the class imbalance identified in the EDA, accuracy was treated as a secondary metric, with primary emphasis placed on specificity and MCC to assess the model’s predictive utility for the minority class.

3 Results and Discussion

3.1 Discussion and Future Work

The Duke Breast Cancer dataset was analyzed to evaluate whether radiomic phenotypes carry a detectable signal for Homologous Recombination Deficiency

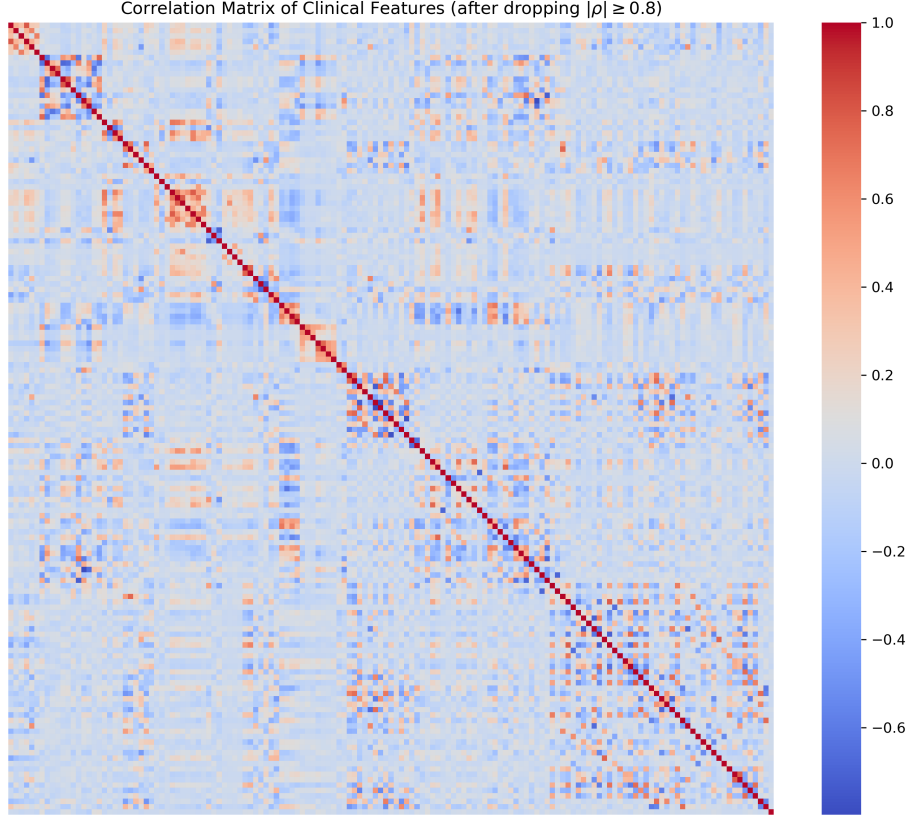


Figure 1: Pearson absolute correlation matrix of the initial imaging feature set. Highly correlated features ($|\rho| \geq 0.8$) were identified from this matrix and pruned to mitigate multicollinearity, thereby reducing model overfitting and ensuring the interpretability of the subsequent feature importance analysis.

Table 1: Baseline Random Forest Classification Performance (10-Fold Cross-Validation)

Metric	Mean $\pm$ SD
Accuracy	81.0% $\pm$ 1.1%
Sensitivity	3.2% $\pm$ 3.2%
Specificity	98.4% $\pm$ 1.2%
Matthew’s Corr. Coeff. (MCC)	0.043 $\pm$ 0.096

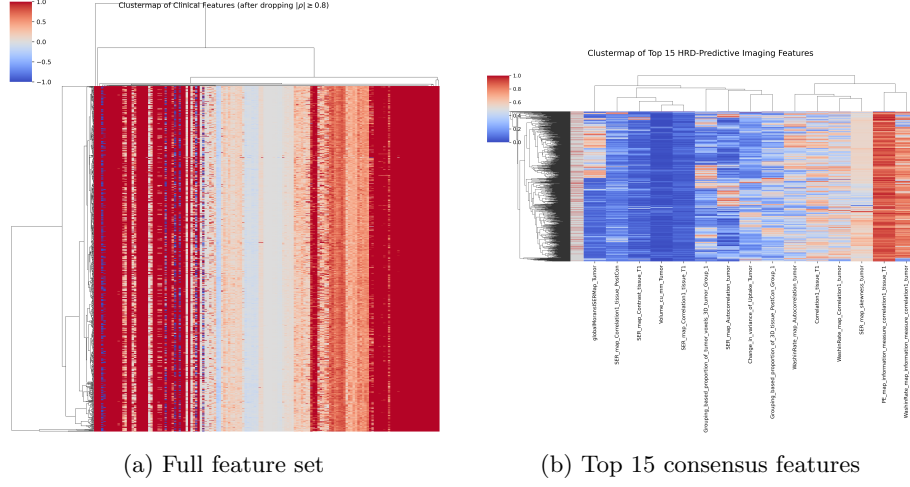


Figure 2: Comparison of unsupervised hierarchical clustering results. (a) The full imaging feature set exhibits high baseline heterogeneity. (b) The top 15 consensus features selected via Random Forest importance demonstrate a refined signature, providing clearer discriminative patterns between the HRD-enriched and non-enriched subtypes.

(HRD). The pipeline initiated with comprehensive feature engineering, incorporating dataset cleaning, multicollinearity mitigation via automated correlation thresholds, and feature selection driven by Gini importance within a Random Forest framework. A Random Forest classifier was selected as the primary modeling architecture due to its capacity to capture complex, non-linear relationships without relying on rigid parametric assumptions regarding the underlying data distribution (in contrast to traditional linear approaches like logistic regression).

While the baseline model demonstrated high overall accuracy (81.0%) and exceptional specificity (98.4%), its critically low sensitivity (3.2%) underscores a classic class-imbalance bottleneck. To move past this baseline performance, several clear avenues for future work are proposed:

- **Advanced Statistical Filtering:** Integrate formal univariate hypothesis testing (such as Student’s t-test or the Mann-Whitney U test) to isolate features exhibiting statistically significant differences between cohorts prior to ensemble modeling. Furthermore, calculating individual receiver operating characteristic (ROC) curves for each feature would extend the depth of the current multi-feature pipeline.
- **Model Interpretability Frameworks:** Implement post-hoc explainability methods such as SHAP (SHapley Additive exPlanations) to explicitly quantify, rank, and visualize the directional impact and contribution of individual radiomic features on the classifier’s predictions.

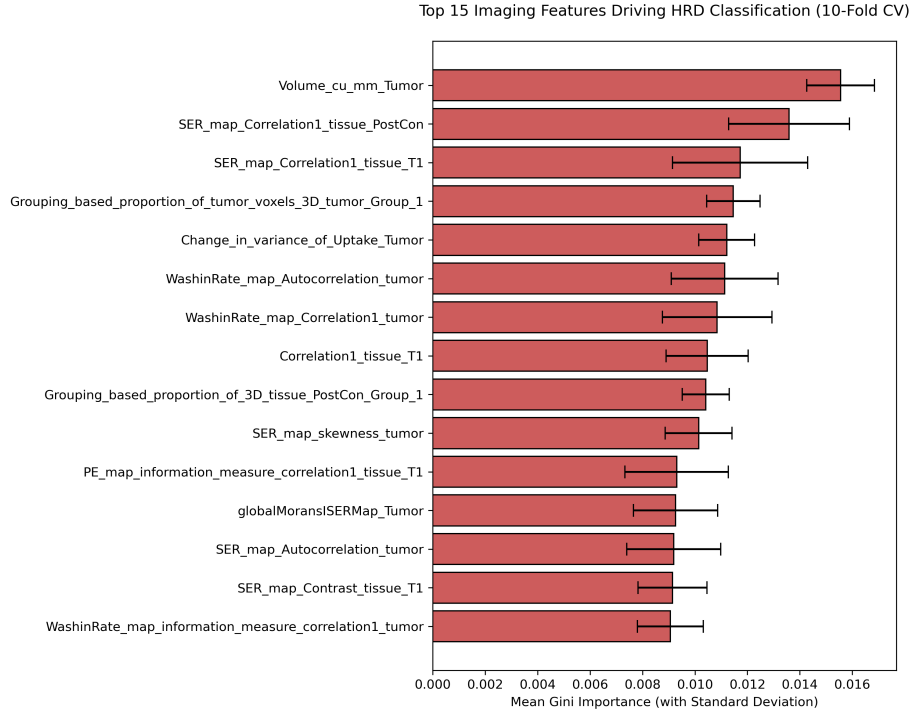


Figure 3: Top 15 consensus imaging features ranked by mean Gini importance. Importances were derived from a Random Forest classifier across 10-fold stratified cross-validation. Error bars represent the standard deviation of importance scores across folds, demonstrating the stability and reproducibility of these features in distinguishing the HRD proxy phenotype.

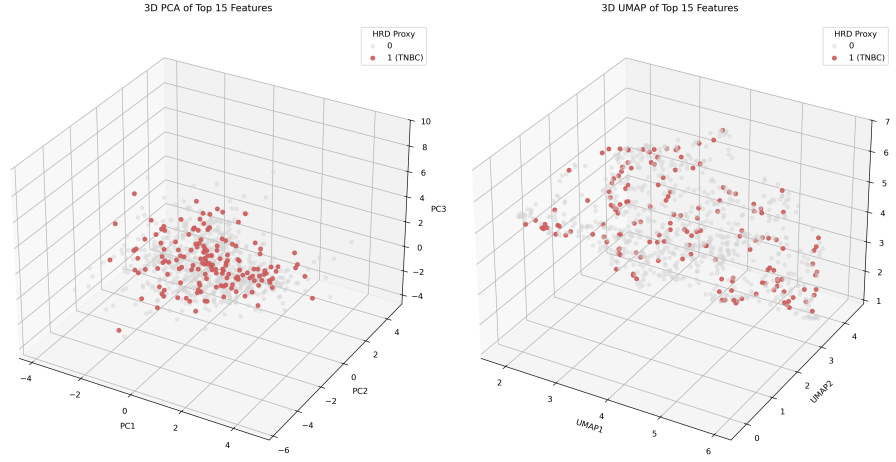


Figure 4: Dimensionality reduction of the top 15 radiomic features using 3D Principal Component Analysis (left) and 3D Uniform Manifold Approximation and Projection (right). The high degree of cluster overlap for HRD-enriched (red) vs. non-enriched (gray) subtypes illustrates the non-linear complexity of the radiomic signal, providing a technical justification for transitioning from handcrafted feature-based modeling to hierarchical Deep Learning frameworks in future work.

- **Hyperparameter Optimization and Class Balancing:** Conduct a systematic hyperparameter sweep (optimizing tree depth, number of estimators, and split criteria) alongside cost-sensitive class weighting or synthetic oversampling (e.g., SMOTE) to explicitly penalize minority class misclassifications and correct the performance skew.
- **Transition to Deep Learning Architectures:** Given that the 3D PCA and UMAP projections confirm a highly non-linear feature space with high cluster overlap, handcrafted radiomic features may be insufficient to fully decouple the biological signal. Future work will explore end-to-end deep learning frameworks capable of automatically extracting hierarchical spatial features. This includes evaluating 1D neural networks on extracted feature profiles, 2D convolutional neural networks (CNNs) using selected MRI slices paired with region-of-interest (ROI) segmentations, or fully 3D CNNs to capture entire volumetric tumor dynamics. Finally, multi-modal learning architectures combining top handcrafted features with raw volumetric MRI sequences could offer a highly comprehensive diagnostic signature.

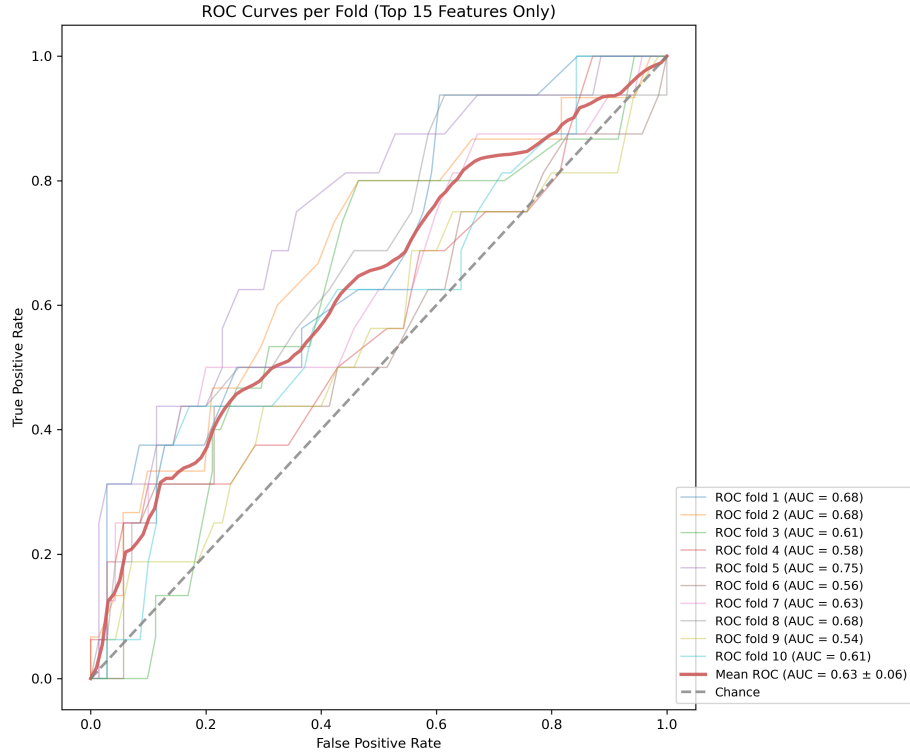


Figure 5: Receiver Operating Characteristic (ROC) curves for the classification of HRD-enriched subtypes using the top 15 consensus imaging features. Curves are plotted for each of the 10 cross-validation folds (light lines), with the bold red line indicating the mean performance. A mean AUC of 0.63 ± 0.06 establishes a robust baseline signal despite the inherent class imbalance of the cohort.